Jaloliddin Ismailov

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Education

Eötvös Loránd University (ELTE) **Budapest, HU**
BSc in Computer Science (GPA: 4.6/5.0; Member of Central Asian Society) *Expected: Jun 2028*

Experience

BME Solar Boat Team **Budapest, HU**
Firmware & Embedded Systems Engineer *Feb 2026 – Present*

- Writing the firmware that controls the boat's throttle on an ESP32, in C and FreeRTOS, and tuning the low-level I2C/SPI drivers so the controls respond instantly and never freeze up.
- Built the system that collects and combines the boat's sensor readings, with bit-level error checking and raw ADC values scaled into numbers that actually mean something. This lets the team watch the boat live and catch failures in the lab through Hardware-in-the-Loop (HIL) testing, instead of finding out on the water.

CERN **Remote / Geneva, CH**
Software Engineering Research Fellow (LHCb) *Nov 2025 – Jan 2026*

- Wrote the algorithms that reconstruct muon tracks through the LHCb detector, in C++ and CUDA so they run on the GPU. The detector throws out a firehose of data, so the whole job was keeping the processing fast enough to not fall behind it.
- Automated the team's build and deploy process with Docker and CI/CD pipelines, so the same code runs the same way everywhere. Mostly this meant people stopped losing hours to manual setup and "works on my machine" problems, and could ship updates faster.

Projects

Assistive Smart Glasses **BME Robotics Hackathon**
Hardware & Edge AI Developer [*Video Demo*] *Jun 2026*

- Built smart glasses that help the wearer navigate, running computer vision and TinyML models directly on a tiny ESP32-CAM (C++ and PlatformIO). Doing the processing on the device instead of sending video off to a server means it works with no internet and reacts instantly – which matters a lot when someone's relying on it to move around safely.

Flood-Risk Monitoring System **CASSINI Hackathon Budapest**
Backend Developer [*Video Demo*] *Apr 2026*

- Built the backend that turns raw environmental sensor data into flood-risk maps, using Python, GeoPandas, and REST APIs. The point was taking numbers that mean nothing to most people and turning them into a map you can glance at and immediately see where the risk is.

Extracurriculars

European Young Engineers **Remote**
Regional Coordinator *Mar 2026 – Present*

- Looking at how machine learning actually gets used alongside climate policy across Hungary, Austria, and Slovakia.

FLEX Program (U.S. Department of State) **Minnesota, USA**
Exchange Student at Prior Lake High School *Sep 2023 – Mar 2024*

- Did 50+ hours of community service and worked with the International Students' Club, mostly bringing student from very different backgrounds together and getting them talking.